Explainer: Bonus Problem 1 (Rudin 7.20) — Approximating $|x|$ by Polynomials

What's going on?

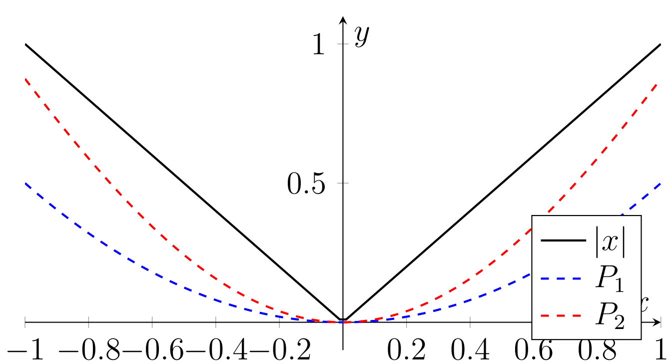
We define polynomials recursively:

$$P_0(x) = 0, \qquad P_{n+1}(x) = P_n(x) + \frac{x^2 - P_n(x)^2}{2},$$

and claim $P_n(x) \to |x|$ uniformly on $[-1, 1]$.

This is remarkable: $|x|$ has a corner at 0 and isn't differentiable there, yet it's the uniform limit of smooth polynomials! (This is a constructive version of the Weierstrass approximation theorem.)

Visualizing the iteration



Each iteration pushes P_n closer to $|x|$ from below. The correction term $\frac{x^2 - P_n^2}{2}$ is always non-negative (since $P_n \leq |x|$), so the sequence is increasing.

The key identity

The hint gives: $|x| - P_{n+1}(x) = \left[|x| - P_n(x)\right] \cdot \left[1 - \frac{|x| + P_n(x)}{2}\right]$.

Let $e_n(x) = |x| - P_n(x)$ be the error. Then:

$$e_{n+1} = e_n \cdot \left(1 - \frac{|x| + P_n}{2}\right).$$

Two things to prove

1. $0 \le P_n \le P_{n+1} \le |x|$ **for** $|x| \le 1$**:**

Since $P_n \le |x| \le 1$, we have $|x| + P_n \le 2$, so the factor $1 - \frac{|x|+P_n}{2} \ge 0$. Combined with $e_n \ge 0$, we get $e_{n+1} \ge 0$, i.e. $P_{n+1} \le |x|$. Also $P_{n+1} - P_n = \frac{x^2 - P_n^2}{2} = \frac{(|x|-P_n)(|x|+P_n)}{2} \ge 0$.

2. The error goes to 0 **uniformly:**

Since $P_n \ge 0$, the factor $1 - \frac{|x|+P_n}{2} \le 1 - \frac{|x|}{2}$. So:

$$e_n(x) \le |x| \cdot \left(1 - \frac{|x|}{2}\right)^n.$$

Now maximize $g(u) = u(1-u/2)^n$ over $u \in [0,1]$. Calculus (or the bound $(1-u/2)^n \le e^{-nu/2}$ and optimizing $ue^{-nu/2}$) gives $\sup_u g(u) \le \frac{2}{n+1}$.

So $\|P_n - |x|\|_\infty \le \frac{2}{n+1} \to 0$. Uniform convergence!

Why this matters

This gives a direct, constructive proof that $|x|$ can be uniformly approximated by polynomials — without invoking the full Weierstrass theorem. Since $\max(f,g) = \frac{f+g+|f-g|}{2}$, this lets you build up approximations to any continuous function via the Stone–Weierstrass machinery.